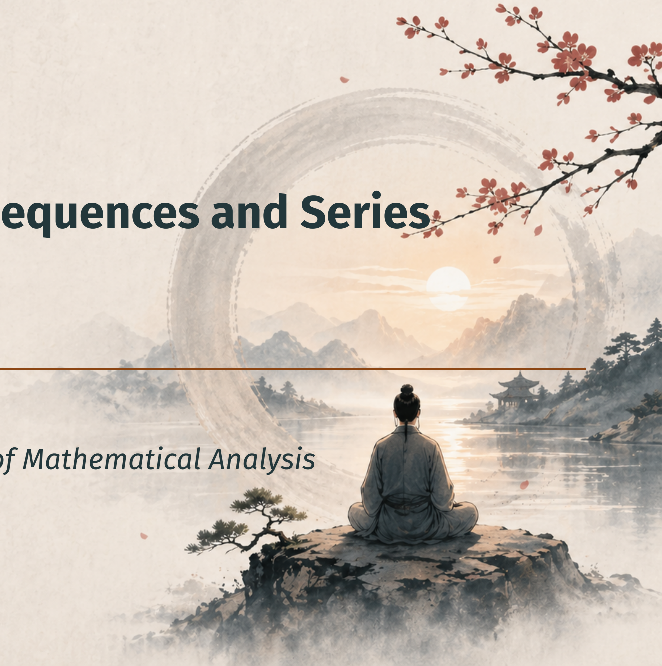
Chapter 3: Numerical Sequences and Series

Section: Subsequences

Based on Walter Rudin's *Principles of Mathematical Analysis*

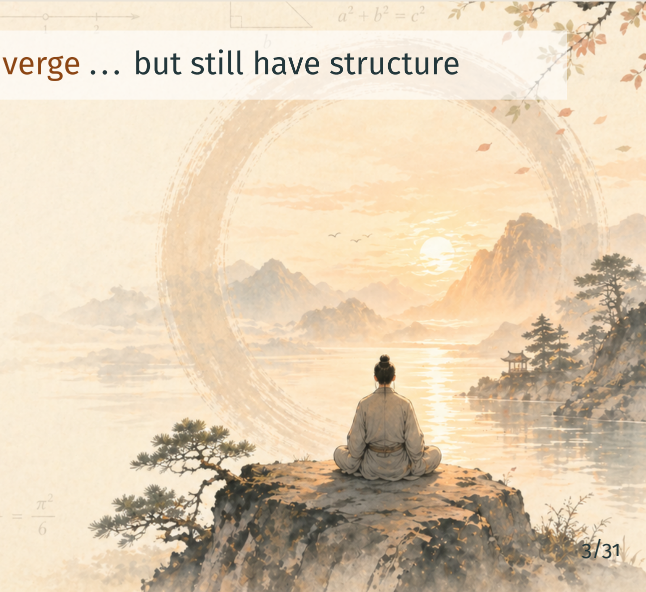


Outline

Motivation

Why Subsequences?

- Many sequences do not converge … but still have structure



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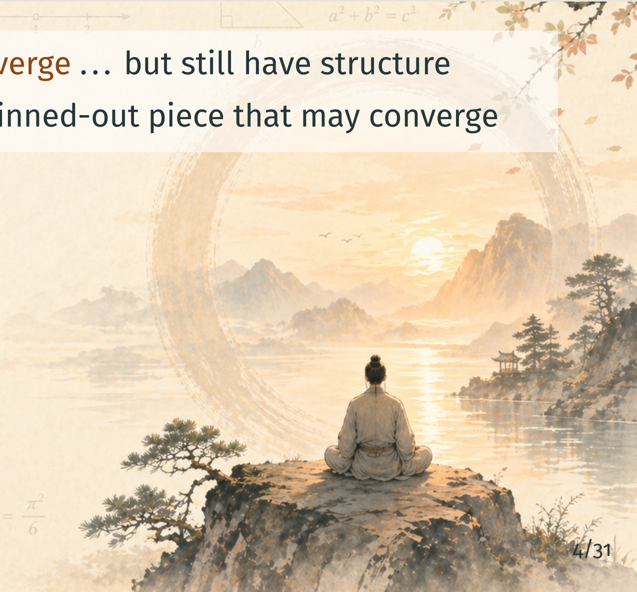
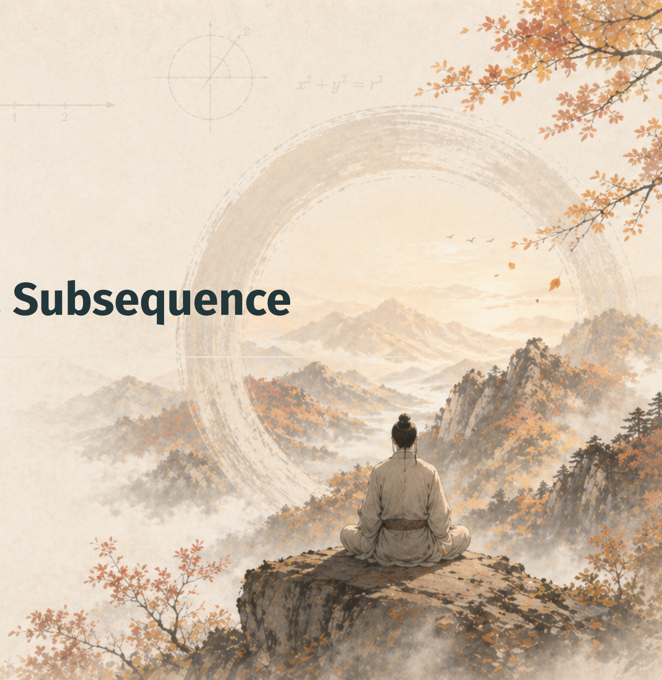
- A subsequence extracts a thinned-out piece that may converge



Why Subsequences?

- Many sequences do not converge … but still have structure
- A subsequence extracts a thinned-out piece that may converge
- In compact spaces, a convergent subsequence *always exists*

Definition of a Subsequence



Definition 3.5: Subsequence

Definition 3.5 — Subsequence

Given a sequence $\{p_n\}$ and a sequence of positive integers

$$n_1 < n_2 < n_3 < \cdots ,$$

the sequence $\{p_{n_k}\}$ is called a subsequence of $\{p_n\}$.

Definition 3.5: Subsequential Limit

Definition 3.5 — Subsequential Limit

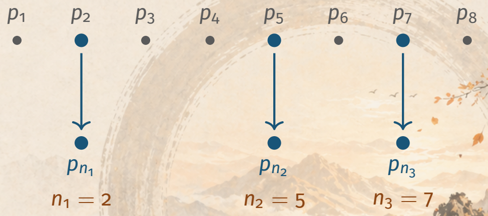
Given $\{p_n\}$ and a subsequence $\{p_{n_k}\}$:

- If $\{p_{n_k}\}$ converges, its limit is a subsequential limit of $\{p_n\}$.
- Write E^* for the set of all subsequential limits.

Picture: A Subsequence Is a Thinned Sequence

Original $\{p_n\}$:

Subsequence $\{p_{n_k}\}$:



Keep some terms (in order), drop the rest.

Example: $p_n = (-1)^n$

Two Subsequential Limits

Let $p_n = (-1)^n$ in $\mathbb{R}$. The sequence $\{p_n\}$ diverges, but:

- Even indices ($n_k = 2k$): $p_{n_k} = 1 \to 1$.
- Odd indices ($n_k = 2k - 1$): $p_{n_k} = -1 \to -1$.

So $E^* = \{-1, +1\}$.

Remark: Convergence and Subsequences

Useful Equivalence

$$p_n \to p \iff \text{every subsequence } \{p_{n_k}\} \text{ converges to } p.$$

Two-way picture:

- $(\Rightarrow)$ Tail of $\{p_n\}$ in ε-ball $\implies$ tail of $\{p_{n_k}\}$ in ε-ball.
- $(\Leftarrow)$ The original $\{p_n\}$ *is itself* a subsequence.

Existence of Convergent Subsequences

Theorem 3.6: Compactness Forces Convergence

Theorem 3.6 (a)

If $\{p_n\}$ is a sequence in a compact metric space X, then some subsequence of $\{p_n\}$ converges to a point of X.

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Theorem 3.6 (a)

If $\{p_n\}$ is in a compact metric space X, then some subsequence converges in X.

Theorem 3.6 (b)

Every bounded sequence in $\mathbb{R}^k$ contains a convergent subsequence.

Proof Strategy: Theorem 3.6 (a)

Goal: Build a convergent subsequence of $\{p_n\}$ inside compact X.

Two cases by the range $E = \{p_n : n \geq 1\}$:

(I) E is finite: some value p is hit infinitely often $\Rightarrow$ constant subsequence.

(II) E is infinite: by Theorem 2.37, E has a limit point $p \in X$. Build a subsequence inductively, using $\varepsilon = 1/k$ at step k.

Proof of 3.6 (a) — Case I: E Finite

Case I: finite range

If $E = \{p_n : n \geq 1\}$ is finite, some value $p \in E$ is taken infinitely often:

$$p_{n_1} = p_{n_2} = p_{n_3} = \cdots = p, \quad n_1 < n_2 < n_3 < \cdots .$$

The constant subsequence $\{p_{n_k}\}$ converges to $p \in X$. $\square$

Proof of 3.6 (a) — Case II: Step 1

Case II: E infinite, Step 1 — find a limit point

By Theorem 2.37, every infinite subset of a compact set has a limit point in the set.

$$\Longrightarrow \qquad E \text{ has a limit point } p \in X.$$

Proof of 3.6 (a) — Case II: Step 2

Step 2 — pick the first index n_1

Since p is a limit point of E, every neighborhood of p contains points of E other than p.

Choose n_1 with

$$d(p, p_{n_1}) < 1.$$

Proof of 3.6 (a) — Case II: Step 3

Step 3 — inductive choice of n_k

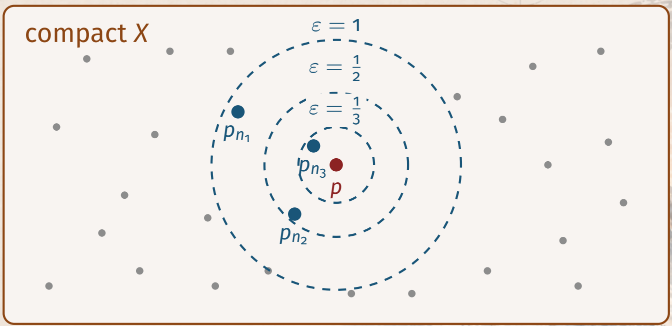
Suppose $n_1 < n_2 < \cdots < n_{k-1}$ have been chosen.

By Theorem 2.20: every neighborhood of a limit point contains *infinitely many* points of E.

So there is $n_k > n_{k-1}$ with

$$d(p, p_{n_k}) < \frac{1}{k}.$$

Picture: Inductive Construction of the Subsequence



Each step shrinks ε to $1/k$ and pulls one new term into the smaller ball.

Proof of 3.6 (a) — Case II: Step 4

Step 4 — the subsequence converges to p

For all $k \geq 1$:

$$d(p, p_{n_k}) < \frac{1}{k}.$$

Given $\varepsilon > 0$, choose $N \geq 1/\varepsilon$. Then for all $k \geq N$:

$$d(p, p_{n_k}) < \frac{1}{k} \leq \frac{1}{N} \leq \varepsilon.$$

Hence $p_{n_k} \to p \in X$. $\square$

Proof of 3.6 (b) — The Bounded Case in $\mathbb{R}^k$

Reduction to part (a)

Let $\{p_n\}$ be a bounded sequence in $\mathbb{R}^k$.

- Theorem 2.41: every bounded subset of $\mathbb{R}^k$ lies in a compact set K.
- So $\{p_n\}$ is a sequence in compact K.
- Apply part (a) $\Longrightarrow$ a subsequence converges in $K \subset \mathbb{R}^k$. $\square$

Subsequential Limits Form a Closed Set

Theorem 3.7: E^* Is Closed

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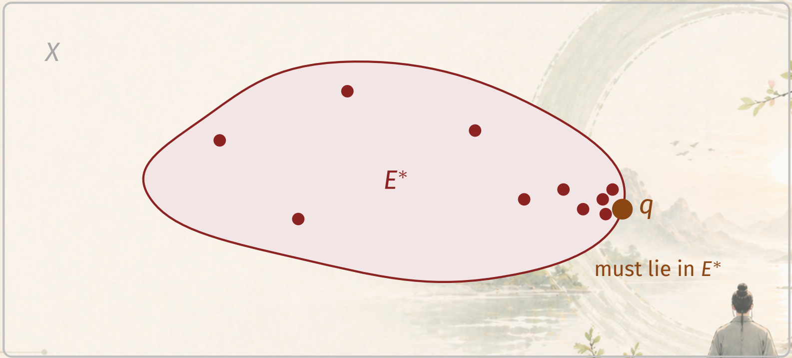
The set E^* of subsequential limits of a sequence $\{p_n\}$ in a metric space X is a closed subset of X.

Why this matters:

- Limits of subsequential limits are again subsequential limits.
- E^* contains all its limit points — nothing escapes.

Picture: E^* as a Closed Set



Proof Strategy: Theorem 3.7

Goal: Show every limit point q of E^* lies in E^*.

Approach: Build a subsequence $\{p_{n_k}\}$ of $\{p_n\}$ with $p_{n_k} \to q$.

Trick: Use both facts at each step:

- "q is a limit point of E^*" $\implies$ pick $x \in E^*$ near q.
- "$x \in E^*$" $\implies$ pick p_{n_k} near x.

Triangle inequality then forces p_{n_k} near q.

Proof of 3.7 — Step 1: Setup

Step 1 — the construction plan

Let q be a limit point of E^*.

Set $n_0 = 0$ (so that "$n_k > n_{k-1}$" makes sense at $k = 1$).

We will build $0 < n_1 < n_2 < n_3 < \cdots$ inductively so that

$$d(q, p_{n_k}) \to 0 \quad \text{as } k \to \infty.$$

Proof of 3.7 — Step 2: Inductive Choice

Step 2 — inductive choice of n_k

Suppose $n_1, \ldots, n_{k-1}$ are chosen. Pick:

- $x \in E^*$ with $d(x, q) < \dfrac{1}{k}$ (possible: q is a limit point of E^*).
- $n_k > n_{k-1}$ with $d(x, p_{n_k}) < \dfrac{1}{k}$ (possible: $x \in E^*$).

Proof of 3.7 — Step 3: Triangle Inequality

Step 3 — estimate $d(q, p_{n_k})$

By the triangle inequality:

$$d(q, p_{n_k}) \le d(q, x) + d(x, p_{n_k}) < \frac{1}{k} + \frac{1}{k} = \frac{2}{k}.$$

Hence

$$\boxed{d(q, p_{n_k}) < \frac{2}{k} \quad \text{for all } k \ge 1.}$$

Proof of 3.7 — Step 4: Conclusion

Step 4 — $\{p_{n_k}\}$ converges to q

Since $2/k \to 0$ as $k \to \infty$:

$$d(q, p_{n_k}) \to 0 \quad \Longrightarrow \quad p_{n_k} \to q.$$

So $q \in E^*$.

Hence E^* contains all its limit points: E^* is closed. $\square$

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Key takeaways:

- Definition 3.5: subsequence $\{p_{n_k}\}$ via strictly increasing indices.
- Convergence test: $p_n \to p$ iff every subsequence converges to p.
- Theorem 3.6: compact $\Rightarrow$ a convergent subsequence exists; bounded in $\mathbb{R}^k \Rightarrow$ same.
- Theorem 3.7: the set E^* of subsequential limits is closed.

Coming up next: Cauchy sequences.